

Unit 1: Introduction to Research, Ethics & Literature Review

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Introduction: Meaning of research, objectives, and scientific voyage Types of Research: Descriptive vs. Analytical, Applied vs. Fundamental, Quantitative vs. Qualitative Ethics in Research: Core ethical values, institutional integrit...

Table of Contents

1. Meaning & Nature of Research	p. 2
2. Objectives & Motivations in Research	p. 3
3. Types of Research	p. 4
4. Ethics in Research	p. 5
5. Types of Research Misconduct (The FFP Triad & Beyond)	p. 6
6. Literature Review & Analysis of Prior Art	p. 7
7. Critical and Creative Reading in Research	p. 8
8. Bibliographic Databases & Metrics (hhh-index, JIF)	p. 8
9. Citations, Knowledge Flow & Attributions	p. 10
10. Solved Problems & Bibliometric Calculations	p. 12
11. Quick Revision Sheet	p. 15
Authentic Exam Archive & Model Solutions	p. 16
13. Prescribed References	p. 23

1. Meaning & Nature of Research

Definition:

Research is a systematic, organized, and objective investigation undertaken to establish facts, solve problems, test hypotheses, or discover new knowledge through the rigorous application of the scientific method.

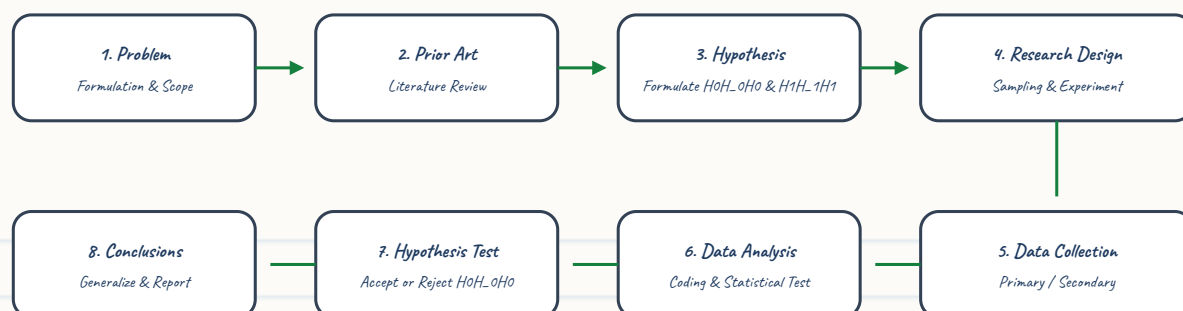
Etymologically, research comprises two Latin-derived parts: "**Re**" (meaning again, anew, or afresh) and "**Search**" (meaning to examine closely and carefully, or to test and try). It is fundamentally a movement from the known to the unknown—a voyage of discovery.

1.1 Research vs. Random Exploration

Unsystematic trial-and-error differs drastically from scientific research. Research demands:

- **Systematization:** Planned, structured execution following established methodological protocols.
- **Empirical Evidence:** Reliance on observable, measurable phenomena rather than superstition, tradition, or unverified claims.
- **Reproducibility (Replicability):** Independent researchers following identical procedures under equivalent constraints must obtain commensurate findings.
- **Controlled Conditions:** Isolating causal relationships by moderating extraneous variables.

Figure 1.1: The Sequential Stages of the Scientific Research Process



Steps in research as outlined in C.R. Kothari's methodology framework. Note that feedback loops occur between analysis, hypothesis refinement, and design.

Quick Recall:

Research is a scientific voyage moving systematically from known axioms to unknown realities via hypothesis formulation, experimental control, empirical evidence, and peer validation.

2. Objectives & Motivations in Research

The principal objective of research is to unearth answers to questions through scientific procedures that prevent subjective human bias from corrupting the truth.

2.1 Four Broad Classes of Research Objectives

1. **Exploratory / Formulative:** To achieve familiarity with an unfamiliar phenomenon or achieve new insights (e.g., preliminary pilot study into an emerging zero-day cybersecurity vulnerability).
2. **Descriptive:** To accurately portray the characteristics of a particular individual, situation, community, or software architecture (e.g., census surveys, measuring average developer turnover rate in Bangalore tech firms).
3. **Diagnostic:** To establish the frequency with which an event occurs or with which it associates with another condition (e.g., investigating why database latency spikes sharply during flash sales).
4. **Hypothesis-Testing (Causal):** To test a causal hypothesis asserting a definitive relationship between dependent and independent variables (e.g., testing whether increasing CPU cache from 16MB to 32MB significantly reduces inference latency of a transformer model).

2.2 What Motivates People to Undertake Research?

- **Intellectual Curiosity:** The desire to understand the unexplained mechanics of natural or computational phenomena.
- **Solving Practical Challenges:** The urge to resolve societal, environmental, or engineering bottlenecks (e.g., developing low-power IoT MAC protocols).
- **Academic & Professional Growth:** Earning an advanced research degree (M.S., Ph.D.) or gaining intellectual recognition among peers.
- **Social Responsibility & Utility:** Directing technical innovation to elevate human well-being and public safety.

Quick Recall:

Four objective pillars: Exploratory (new insights), Descriptive (profiling traits), Diagnostic (frequency & association), and Hypothesis-testing (causal verification).

3. Types of Research

In academic examinations (such as Make-Up April 2025 and SEE Feb/March 2025), questions regularly demand clear tabular contrasts between the major research paradigms.

Classification Dimension	Paradigm A	Paradigm B	Core Distinction
Inquiry Purpose	Descriptive Research	Analytical Research	Descriptive reports "what is" (ex post facto facts without variable control); Analytical evaluates "why and how" using critical causal analysis.
Utility & Scope	Fundamental / Basic / Pure	Applied Research	Fundamental seeks knowledge expansion and general theory formulation (e.g. Einstein's relativity, P vs NP); Applied targets immediate practical solution to a pressing industry/societal problem.
Data Modality	Quantitative Research	Qualitative Research	Quantitative measures phenomena numerically (metrics, ratios, statistical tests); Qualitative explores subjective meanings, motivations, attitudes, and behavioral contexts.
Methodological Basis	Conceptual Research	Empirical Research	Conceptual relates to abstract ideas, philosophies, or reinterpreting existing concepts (favored by philosophers and theorists); Empirical relies on hard observation and verifiable experimental experiments.

Classification Dimension	Paradigm A	Paradigm B	Core Distinction
Time Horizon	Cross-Sectional / One-Time	Longitudinal Research	Cross-sectional captures data across subjects at a single snapshot in time; Longitudinal observes subjects repeatedly over extended temporal periods.

Common Mistake:

Confusing Applied Research with Empirical Research. Applied research refers to the intent (solving an immediate industry problem), whereas Empirical refers to the method (data collection via experimental observation). An applied project can be empirical, but so can a fundamental scientific experiment!

Quick Recall:

Key pairs to memorize: Descriptive (describes status quo) vs. Analytical (proves causality); Fundamental (theory expansion) vs. Applied (real-world problem solving); Quantitative (numerical measurement) vs. Qualitative (contextual meaning).

4. Ethics in Research

Research ethics provides the moral compass governing all scholarly inquiry, data curation, experimental treatment, and publication dissemination.

4.1 Foundational Ethical Principles

- **Honesty:** Honestly report data, results, methods, and publication status. Never deceive colleagues, grant sponsors, or the public.
- **Objectivity:** Eliminate bias in experimental design, data interpretation, peer review, personnel decisions, and expert testimony.
- **Integrity:** Keep agreements and promises; act with sincerity; strive for consistency of thought and action.
- **Carefulness:** Avoid careless errors and negligence; examine your work and colleagues' work with rigorous scrutiny; maintain spotless logs.
- **Openness:** Share data, results, tools, resources, and ideas openly. Remain receptive to novel ideas and healthy criticism.

- **Respect for Intellectual Property:** Honor patents, copyrights, trademarks, trade secrets, and licenses. Never use unpublished data without prior authorization. Always give credit where credit is due.
- **Confidentiality:** Safeguard confidential communications (e.g., papers submitted for blind peer review, grant proposals, patient health records, industrial trade secrets).
- **Social Responsibility:** Strive to promote social good and prevent harm through research, public education, and advocacy.

Quick Recall:
Ethical research rests on the bedrock of Honesty, Objectivity, Carefulness, Openness, and unconditional respect for Intellectual Property.

5. Types of Research Misconduct (The FFP Triad & Beyond)

Core Triad:
In international scientific consensus (including US Office of Research Integrity and Indian UGC/AICTE regulations), research misconduct is explicitly defined as **Fabrication**, **Falsification**, or **Plagiarism** (FFP) in proposing, performing, or reviewing research, or in reporting research results.

Misconduct Type	Formal Definition	Concrete Engineering Example
Fabrication	Inventing data, synthetic observations, or fictitious experiments and recording/reporting them as genuine real-world trials.	Simulating 1,000 sensor battery runs in an Excel sheet by typing random discharge numbers instead of running physical field tests.
Falsification	Manipulating research materials, equipment, or processes, or changing/omitting data points such that research is not accurately represented in record.	Discarding 14 latency outliers because they violate an algorithm's claimed sub-millisecond execution bound.
Plagiarism	Appropriation of another person's ideas, processes, results, code, or text without giving appropriate attribution or credit.	Pasting a competitor's neural network architecture and mathematical formulation into

5.2 Secondary Forms of Ethical Malpractice

- **Self-Plagiarism (Duplicate Publication):** Re-publishing one's own previously copyrighted text, figures, or findings in a new venue without clear disclosure or citation.
- **Salami Slicing (Data Fragmentation):** Partitioning a coherent, unified large study into thin, minimally publishable units solely to artificially inflate publication counts.
- **Gift / Honorary Authorship:** Adding the name of a senior professor, department chair, or influential acquaintance to an author list when they contributed no intellectual work.
- **Ghost Authorship:** Concealing the true author (e.g., an industrial ghostwriter hired by a pharmaceutical firm) to mask bias.
- **Predatory Publishing:** Submitting manuscripts to commercial pay-to-publish scam journals that bypass authentic peer review for financial gain.

Quick Recall:

FFP Triad: Fabrication = making up data; Falsification = altering data/dropping outliers; Plagiarism = stealing another's work without attribution.

6. Literature Review & Analysis of Prior Art

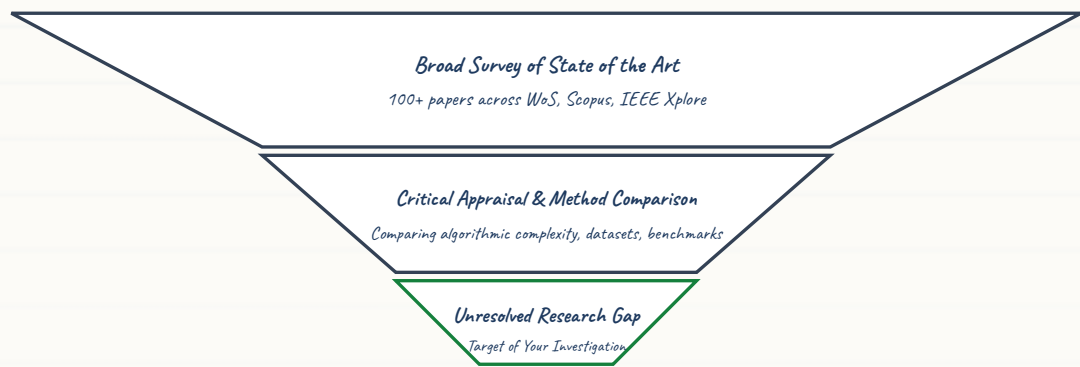
A literature review is not a superficial bibliography; it is an analytical synthesis of published scholarship establishing the context and justification for your investigation.

6.1 Interplay of New and Existing Knowledge

Scientific advancement is cumulative. Sir Isaac Newton famously observed: "If I have seen further, it is by standing on the shoulders of Giants."

- **Grounding:** Existing knowledge provides validated methodologies, established theoretical models, and standard benchmarks.
- **De-duplication:** Prevents "reinventing the wheel" by ensuring researchers do not replicate solved problems.
- **Gap Identification:** Highlighting limitations, contradictory findings, unexamined boundary conditions, or scalability bottlenecks in the existing literature.
- **Synthesizing:** Connecting disjoint findings from distinct disciplines to construct a novel overarching framework.

Figure 1.2: The Literature Review Analytical Synthesis Funnel



The literature synthesis funnel: filtering broad bibliographies down to actionable, verifiable research questions.

Quick Recall:

A literature review identifies prior art, avoids redundant wheel-reinvention, and isolates the precise research gap that justifies your novel study.

7. Critical and Creative Reading in Research

Technical reading demands a higher cognitive engagement than casual reading. Researchers must read on two intertwined planes: *critically* and *creatively*.

Attribute	Critical Reading (Analytical Lens)	Creative Reading (Generative Lens)
Primary Mindset	Skeptical, evaluating, dissecting validity and soundness.	Visionary, connecting dots, exploring possibilities.
Guiding Questions	"Are the assumptions justified? Are the statistics rigorous? Does the conclusion truly follow from the evidence?"	"What if we relaxed assumption X? Can this technique be applied to an entirely different engineering domain?"
Focus Areas	Threats to validity, unstated biases, confounding variables.	Untapped cross-pollinations, novel architectural hybridizations.
Outcome	Accurate judgment of the paper's actual scientific contribution.	Conceptualizing your own innovative research hypothesis.

Quick Recall:

Critical reading validates the soundness and rigor of an author's claims; creative reading

8. Bibliographic Databases & Metrics (hhh-index, JIF)

Reliable scholarly literature must be discovered through curated, indexed bibliographic repositories rather than generic commercial search engines.

8.1 Leading Bibliographic Repositories

- **Web of Science (WoS):** Maintained by Clarivate Analytics; the oldest citation indexing service (originating from Eugene Garfield's Science Citation Index). Highly curated, covers high-impact journals, calculates the official Journal Citation Reports (JCR) Impact Factor.
- **Scopus:** Maintained by Elsevier; comprehensive multidisciplinary abstract and citation database indexing over 25,000 peer-reviewed journals, book series, and conference proceedings. Generates CiteScore and SJR metrics.
- **IEEE Xplore:** Digital library dedicated to electrical engineering, electronics, computer science, and telecommunications. High relevance for engineering students.
- **Google Scholar:** Free web search engine indexing scholarly literature across publishers. Highly comprehensive but includes uncurated, pre-print, and predatory sources without quality filtering.

8.2 Quantitative Bibliometrics: Definitions & Formulas

Formula 1: The Hirsch Index (hhh-index):

A researcher has index h if h of their N_p papers have at least h citations each, and the other $(N_p - h)$ papers have $\leq h$ citations each.

$$\max\{r \mid c_r \geq r\}$$

where papers are sorted in descending order of citation counts $c_1 \geq c_2 \geq \dots \geq c_{N_p}$, and r is the 1-indexed rank.

Formula 2: The i10i10i10-Index:

Used widely by Google Scholar, the $i10$ -index is simply the number of academic publications by an author that have received at least 10 citations each:

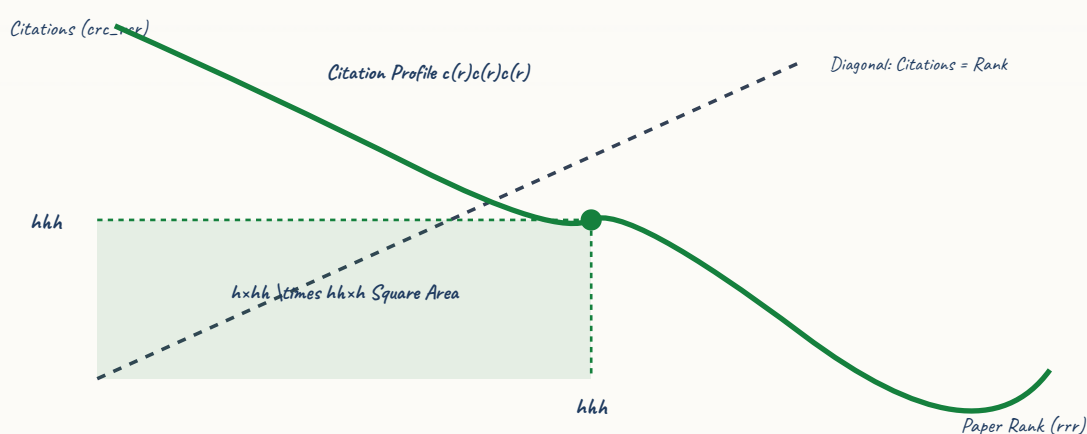
$$i10 = \sum_{i=1}^{N_p} \mathbb{I}(c_i \geq 10)$$

Formula 3: Two-Year Journal Impact Factor (JIF):

The Journal Impact Factor for year t measures the average frequency with which peer-reviewed articles published in that journal during the preceding two years were cited in year t :

$$\text{JIF}_t = \frac{\text{Citations in year } t \text{ to items published in } (t - 1) \text{ and } (t - 2)}{\text{Total citable items (articles + reviews) published in } (t - 1) \text{ and } (t - 2)}$$

Figure 1.3: Graphical Determination of the h -index



The h -index corresponds to the maximum square that fits under the sorted citation curve, balancing productivity with scientific impact.

Quick Recall:

h -index = maximum rank r where paper citations $\geq r$. JIF = citations in year t to articles published in previous 2 years divided by total citable items published in those 2 years.

9. Citations, Knowledge Flow & Attributions

A citation is a formal reference linking an author's claim to an earlier published work, forming the connective tissue of human intellectual heritage.

9.1 Functions and Attributes of Citations

- **Verification & Evidence:** Allows critical peers to audit the empirical validity and context of claims.
- **Intellectual Property & Priority:** Acknowledges who first conceived an idea, established

- **Navigational Roadmap:** Directs readers toward deeper technical treatments and foundational proofs.
- **Defensive Intellectual Property:** Demonstrates prior art when defending against invalid patent filings.

9.2 Impact of Title and Keywords on Citation Velocity

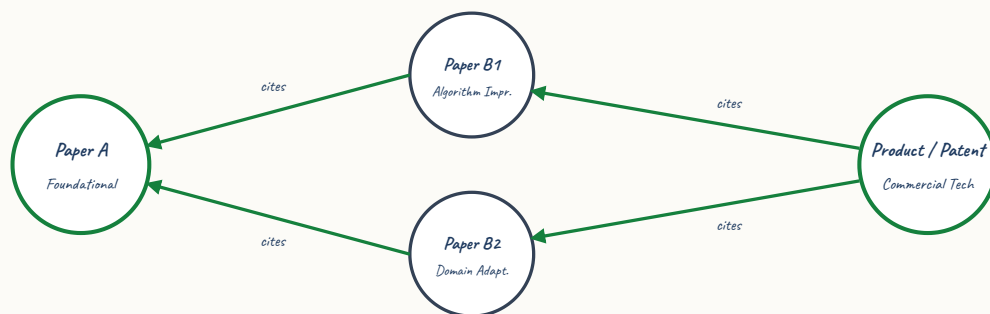
In modern automated indexing systems, the title and author keywords dictate discovery:

- **Title Specificity:** Concise, declarative, and informative titles attract significantly higher click-through and download rates than vague or poetic phrasing.
- **Keywords Optimization:** Choosing standard MeSH / IEEE taxonomy keywords ensures that search algorithms surface the manuscript in automated queries.
- **Accessibility Factor:** If indexing spiders fail to match metadata to active researcher searches, even groundbreaking theorems remain uncited.

9.3 Knowledge Flow Through Citation Networks

Knowledge diffuses across geography and academic disciplines via directed citation graphs. When Paper *C* cites Paper *B*, which cited foundational Paper *A*, intellectual concepts propagate outward, spawning commercial patents, technical standards, and follow-on breakthroughs.

Figure 1.4: Directed Acyclic Graph of Knowledge Diffusion via Citations



Citations flow backwards to acknowledge intellectual origins; knowledge flows forward to drive new implementations.

9.4 Major Citation Styles: IEEE vs. APA

Element	IEEE Style (Numbered Citation)	APA Style (Author-Date Citation)
In-text Citation	Bracketed sequential integer: [1], [2] - [4].	Author and year in parentheses: (Kothari, 2019) or Kothari (2019).
Journal Article Entry	A. B. Author, "Title of article," <u>Abbrev. Title of Journal</u> , vol. x, no. y, pp. 12-25, Month, Year.	Author, A. A. (Year). Title of article. <u>Title of Periodical</u> , x(y), 12-25.
Primary Field	Engineering, Computer Science, Telecommunications.	Psychology, Social Sciences, Education, Business.

Quick Recall:

Citations provide evidentiary tracing and IP priority. IEEE uses bracketed integers [1]; APA uses author-year (Kothari, 2019).

10. Solved Problems & Bibliometric Calculations

Rank (r)	Citations (C_r)	Condition: $C_r \geq r$?	Status
1	42	$42 \geq 1$	Satisfied
2	28	$28 \geq 2$	Satisfied
3	17	$17 \geq 3$	Satisfied
4	14	$14 \geq 4$	Satisfied
5	11	$11 \geq 5$	Satisfied
6	8	$8 \geq 6$	Satisfied
7	7	$7 \geq 7$	Satisfied (boundary)
8	5	$5 \geq 8$	Violated ($5 < 8$)
9	2	$2 \geq 9$	Violated
10	1	$1 \geq 10$	Violated

Solved Problem 1.2: Computing 2-Year Journal Impact Factor (JIF):

Problem Statement: A computing journal publishes research articles and reviews. The publication and citation records for the past three years are:

- Year 2022: Published 120 research articles and review papers.
- Year 2023: Published 140 research articles and review papers.
- During Year 2024:
 - Papers published in 2022 received 650 citations.
 - Papers published in 2023 received 780 citations.

Calculate the official 2024 Journal Impact Factor (JIF) of the journal.

Step 1: Identify total citable items published in 2022 and 2023 (D):

$$D = \text{Items}(2022) + \text{Items}(2023) = 120 + 140 = 260$$

Step 2: Identify total citations received in 2024 to items published in 2022 and 2023 (N):

$$N = \text{Citations in 2024 to 2022 items} + \text{Citations in 2024 to 2023 items} = 650 + 780 = 1430$$

Step 3: Apply the JIF formula:

$$\text{JIF}_{2024} = \frac{N}{D} = \frac{1430}{260}$$

Carrying out the arithmetic division:

$$\frac{1430}{260} = \frac{143}{26} = 5.500$$

$$\text{JIF}_{2024} = 5.500$$

Interpretation: On average, each citable paper published in this journal during 2022 and 2023 was cited 5.5 times in 2024.

INTERACTIVE BIBLIOMETRICS STUDIO

h-Index, i10-Index & Journal Impact Factor (JIF)

h-Index
Calculator

Journal Impact
Factor

Calculate Jorge E. Hirsch's *h*-index and Google Scholar's *i10*-index from an author's citation distribution:

$$h = \max\{r \mid c_r \geq r\}$$

Paper Citation Counts (comma-separated):

42, 28, 17, 14, 11, 8, 7, 5, 2, 1

Presets:

Dr. Sharma (10 papers, h=7)

Senior Scholar (12 papers, h=8)

Early Career (5 papers, h=3)

H-INDEX

7

7 papers with ≥ 7 cites

I10-INDEX

5

≥ 10 citations each

TOTAL PAPERS

10

In analyzed profile

TOTAL CITATIONS

135

Cumulative impact

Rank (<i>r</i>)	Citations (<i>c_r</i>)	Condition: <i>c_r</i> ≥ <i>r</i> ?	Status
-------------------	------------------------------------	--	--------

INTERACTIVE BIBLIOMETRICS STUDIO

h-Index, i10-Index & Journal Impact Factor (JIF)

h-Index
Calculator

Journal Impact
Factor

Calculate Jorge E. Hirsch's *h*-index and Google Scholar's *i10*-index from an author's citation distribution:

$$h = \max\{r \mid c_r \geq r\}$$

Paper Citation Counts (comma-separated):

42, 28, 17, 14, 11, 8, 7, 5, 2, 1

Presets:

Dr. Sharma (10 papers, h=7)

Senior Scholar (12 papers, h=8)

Early Career (5 papers, h=3)

H-INDEX

7

7 papers with ≥ 7 cites

I10-INDEX

5

≥ 10 citations each

TOTAL PAPERS

10

In analyzed profile

TOTAL CITATIONS

135

Cumulative impact

Rank (<i>r</i>)	Citations (<i>c_r</i>)	Condition: <i>c_r</i> ≥ <i>r</i> ?	Status
-------------------	------------------------------------	--	--------

11. Quick Revision Sheet

Concept	Core Formula / Principle	Exam Trap / Key Insight
Research Definition	Systematic voyage from known to unknown via empirical testing.	Must cite <u>scientific method</u> , <u>replicability</u> , and <u>control</u> .
Research Types	Descriptive (what) vs Analytical (why); Basic (theory) vs Applied (practical).	Descriptive researcher has NO control over variables; ex post facto reporting.
Research Misconduct	FFP = Fabrication (creating data) + Falsification (altering data) + Plagiarism (theft without credit).	Salami slicing = fragmenting one study into multiple small papers.
<i>h</i>-Index	$\max\{r \mid c_r \geq r\}$ when sorted descending.	Ranks must be checked sequentially; once $c_r < r$, the previous rank is h .
JIF (Impact Factor)	$\text{JIF}_t = \frac{\text{Citations in year } t \text{ to items from } (t-1)+(t-2)}{\text{Total citable items published in } (t-1)+(t-2)}$	Denominator contains ONLY citable items (articles & reviews), not letters or editorials.
Databases	WoS (Clarivate, JCR JIF) vs. Scopus (Elsevier, CiteScore).	Google Scholar is uncured and includes non-peer-reviewed preprints.

Authentic Exam Archive & Model Solutions

ARCHIVE

The following authentic questions have been transcribed directly from official department examination papers in [notes/rmipr/practice/](#):

Ramaiah Autonomous Examination · CIE-1 (Oct 29, 2025) Q1.a [7 Marks]

"Define research and list out its objectives and motivations."

Official Model Solution:

1. **Definition:** Research is an organized, systematic, and empirical investigation undertaken to discover new knowledge, establish facts, or verify theories through the scientific method. It is a systematic voyage from the known to the unknown.

2. Objectives of Research:

- Exploratory / Formulative: To gain familiarity with a novel phenomenon or achieve new insights into it.
- Descriptive: To portray accurately the characteristics of a particular individual, situation, or group.
- Diagnostic: To determine the frequency with which an event occurs or its association with other phenomena.
- Hypothesis-Testing (Causal): To test a causal hypothesis between independent and dependent variables.

3. Motivations in Research:

- Desire to get a research degree along with its consequential career benefits.
- Desire to face intellectual challenges in solving unsolved problems.
- Desire to obtain intellectual joy of doing creative work.
- Desire to render social service and contribute to national/industrial welfare.

Dimension	Type A	Type B	Key Distinction
Purpose & Outcome	Fundamental / Basic: Concerned with generalizations and theory formulation (e.g. pure mathematics, quantum physics).	Applied: Aims at finding an immediate solution for an urgent pressing practical problem (e.g. optimizing cloud latency).	Theory generation vs. practical problem solving.
Control & Scope	Descriptive: Ex post facto reporting of affairs as they exist; researcher has no control over variables.	Analytical: Researcher uses available facts or controlled experiments to analyze causal mechanisms.	"What is happening" vs. "Why is it happening".

Dimension	Type A	Type B	Key Distinction
Measurement	Quantitative: Expressed in numerical quantities, statistical counts, and mathematical metrics.	Qualitative: Concerned with qualitative phenomena (human behavior, motives, opinions, sentiments).	Numerical datasets vs. experiential narratives.
Framework	Conceptual: Related to abstract ideas or theories (philosophers, thinkers reinterpreting concepts).	Empirical: Relies on observable experience and experimental verification; data-based research.	Abstract intellectual reasoning vs. observational data.

Ramaiah Autonomous Examination · CIE-1 (Oct 29, 2025) Q3.a [8 Marks]

"What is a plagiarism? Illustrate different types of plagiarisms."

Official Model Solution:

Plagiarism is the uncredited appropriation, theft, or reproduction of another person's ideas, processes, results, words, or artistic expressions without proper formal citation and attribution, representing them as one's own original contribution.

1. **Direct / Verbatim Plagiarism:** Word-for-word transcription of a section of someone else's text without quotation marks and without attribution. This is the most severe ethical breach.
2. **Mosaic / Patchwriting Plagiarism:** Borrowing phrases or sentences from multiple sources and weaving them together with minor synonym replacements, retaining original sentence structures without quotation marks.
3. **Self-Plagiarism (Text Recycling):** Reusing significant portions of one's own previously published copyright-protected papers in a new manuscript without disclosure or citation.
4. **Accidental / Inadvertent Plagiarism:** Failing to cite sources properly due to careless note-taking, misattributing citations, or paraphrasing too closely without malicious intent (still penalized as misconduct).

Ramaiah Autonomous Examination · SEE (Feb/Mar 2025) Q1.a & Q1.b [12 Marks]

"(a) With relevant flow chart describe the necessary steps involved in research process. [6M]

(b) Distinguish between descriptive and analytical research with example. [6M]"

Official Model Solution:

(a) Research Process: 8-step iterative cycle: (1) Problem Formulation → (2)

Literature Survey → (3) Hypothesis Formulation (H_0 , H_1) → (4) Research Design → (5)

Data Collection → (6) Data Analysis → (7) Hypothesis Testing → (8) Generalization & Report. (Refer to Figure 1.1).

(b) Descriptive vs Analytical: Descriptive research surveys current state without manipulation (e.g. measuring student survey responses on AI tool adoption). Analytical research manipulates variables to establish causal relationships (e.g. controlled A/B testing of compiler optimization flags on binary size).

Ramaiah Autonomous Examination · SEE (Feb/Mar 2025) Q1.c [8 Marks]

"What are the key features of the bibliographic database of the Web of Science (WoS), and how is it commonly used in research?"

Official Model Solution:

- **Curated Core Collection:** Strictly peer-reviewed indexing including SCIE, SSCI, A&HCI, and ESCI.
- **Authoritative Journal Metrics:** Exclusive publisher of Journal Citation Reports (JCR) and the official 2-Year Journal Impact Factor (JIF).
- **Bidirectional Citation Navigation:** Tracks cited references (backward prior art) and citing articles (forward knowledge diffusion).
- **Researcher Profiling:** Integrates ResearcherID / Publons to track author h -index, verified peer reviews, and institutional affiliations without identity ambiguity.

Practice Problems with Hidden Step-by-Step Solutions

PRACTICE 1 · h-INDEX COMPUTATION

A researcher has 12 papers with citation counts: [55, 34, 25, 20, 16, 12, 10, 8, 6, 4, 3, 0]. Find their h -index and $i10$ -index.

Practice Problem:

Show Step-by-Step Solution

Solution:

Sort descending and verify $c_r \geq r$:

- Rank 1: 55 (
ge1) • Rank 2: 34 (
ge2) • Rank 3: 25 (
ge3) • Rank 4: 20 (
ge4)
- Rank 5: 16 (
ge5) • Rank 6: 12 (
ge6) • Rank 7: 10 (
ge7) • **Rank 8: 8** (
ge8)
- Rank 9: 6 (< 9) (Condition violated)

h -index = 8, $i10$ -index = 7 (papers with ≥ 10 cites)

Verification: Verified in audit/verify/rmipr/verify_rmipr_u1.py.

PRACTICE 2 · JOURNAL IMPACT FACTOR (JIF)

A journal published 120 papers in 2022 and 140 papers in 2023. In 2024, it received 650 citations to its 2022 papers and 780 citations to its 2023 papers. Calculate its 2024 JIF.

Practice Problem:

Show Step-by-Step Solution

Solution:

$$\text{JIF}_{2024} = \frac{\text{Citations in 2024 to items in 2022 and 2023}}{\text{Citable items published in 2022 and 2023}} = \frac{650 + 780}{120 + 140}$$

Verification: Verified byte-for-byte in
audit/verify/rmipr/verify_rmipr_u1.py.

PRACTICE 3 · FFP MISCONDUCT DISTINCTION

A graduate student alters outlier data points in their dataset so that the p-value falls below 0.05. Is this Fabrication or Falsification? Justify.

Practice Problem:

Show Step-by-Step Solution

Solution:**Falsification.**

Rationale: The student manipulated and altered genuine experimental data points that were actually collected. Had the student invented completely fictitious readings out of thin air, it would constitute Fabrication. Both represent severe research misconduct.

PRACTICE 4 · CITATION NETWORK REASONING

In a citation network, paper X cites paper Y, and paper Y cites paper Z. What does this directed path signify regarding knowledge diffusion?

Practice Problem:

Show Step-by-Step Solution

Solution:

Paper Z is foundational; Paper Y adapted or refined Z's concepts; Paper X applied Y's adapted formulation to a specialized application. Knowledge diffused forward ($Z \rightarrow Y \rightarrow X$), while citation attribution traces backward ($X \rightarrow Y \rightarrow Z$).

PRACTICE 5 · RESEARCH HYPOTHESIS CRITERIA

State the four essential criteria of a valid scientific research hypothesis according to C.R. Kothari.

Practice Problem:

Show Step-by-Step Solution

Solution:

1. **Clear and Precise:** Concepts must be unambiguously defined.
2. **Capable of Verification (Falsifiability):** Must be testable against empirical evidence.
3. **States Relationship Between Variables:** Expresses tentative causal connection between independent and dependent variables.
4. **Consistent with Known Body of Facts:** Aligns with established theoretical knowledge unless explicitly challenging it.

PRACTICE 6 · CITATION INDEXING DATABASES

Compare Scopus and Web of Science in terms of publisher, core metric, and coverage breadth.

Practice Problem:

Show Step-by-Step Solution

Solution:

- **Web of Science:** Clarivate; Journal Impact Factor (JIF); highly selective curation (~21,000 journals).
- **Scopus:** Elsevier; CiteScore, SNIP, SJR; broader coverage (~27,000 journals, extensive conference proceedings).

13. Prescribed References

Official Course Textbooks:

1. **C.R. Kothari and Gourav Garg**, Research Methodology: Methods and Techniques, 4th Edition, New Age International Publishers. (Chapters 1 & 2).
 2. **David V. Thiel**, Research Methods for Engineers, Cambridge University Press, ISBN 978-1-107-03488-4. (Chapters 1, 2 & 3).
 3. **Dipankar Deb, Rajeeb Dey, Valentina E. Balas**, Engineering Research Methodology, Springer Intelligent Systems Reference Library, ISBN 978-981-13-2946-3.
- AI-assisted study notes written from the official syllabus. Verify against the prescribed textbook and your faculty's notes before the exam.